

Shubham Pandey

[Portfolio](#) | [LinkedIn](#) | [GitHub](#) | 8218554651 | contact.shub.aidev@gmail.com

SUMMARY

AI/ML Engineer specializing in Generative AI, Agentic AI, RAG, Machine Learning, Deep Learning, and Computer Vision, with experience building and deploying production-grade full-stack AI systems.

EDUCATION

B.Tech in Computer Science and Engineering (AIML), Bennett University, Greater Noida | 2023-2027 | CGPA: 8.01

TECHNICAL SKILLS

Generative AI & Agentic Systems: LangGraph, LangChain, Agentic Workflows, Multi-Agent Systems, RAG, System Prompting, Semantic Search, Groq API, OpenAI API, Whisper, Prompt Engineering, Vector Database.

Machine Learning, Deep Learning & Computer Vision: PyTorch, TensorFlow, Scikit-Learn, Ultralytics YOLO, Neural Networks, Transfer Learning, OpenCV, MediaPipe, Object Detection, XGBoost, SHAP, Sentence-Transformers.

Backend & Infrastructure: FastAPI, Asynchronous APIs, RESTful Services, Microservices, Pydantic, Uvicorn, WebSockets.

Databases & Developer Tools: FAISS, ChromaDB, Git, GitHub, Postman, Vercel, Miniconda, VS Code, Claude Code, Antigravity, cursor, Trae, Aider, Roo Code. Local LLMs.

EXPERIENCE

AI/ML Engineer — School Management System | Shiksha Bharti Senior Secondary School | June 2026

- Built a RAG-based chatbot to automate parent query resolution, trained on school rules, notices, and exam circulars.
- Developed ML models for early dropout/low-performance detection and fee-defaulter prediction using attendance, academic, and payment data, with results visualized via an interactive analytics dashboard.

PROJECTS

[Interview Coach](#) — *LangGraph, FastAPI, OpenAI, Edge TTS, OpenAI -Whisper, Multi-Agent System*

[GitHub](#)

- Architected an 8-agent LangGraph mock-interview platform generating resume-grounded technical, behavioral, and system-design questions, with adaptive escalation to deeper follow-ups when scores fall below 65%.
- Developed a local Whisper-based speech-analytics pipeline measuring speaking pace (WPM) and filler-word usage, combined with LLM-based answer grading to generate personalized learning plans.

[Finvexis AI](#) — *FastAPI, LangChain, Vector DB, Groq API, Multi-Agent System, Isolated Agents*

[GitHub](#)

- Designed a full-stack, multi-domain AI platform unifying Business Strategy, Finance Automation, and Sales/HR workflows via 12 modular, domain-isolated, independently scalable AI agents.
- Implemented multi-agent system architecture enabling independent scaling, orchestration, and coordination of forecasting, invoicing, and sales-assistance capabilities across business domains.

[PolyDoc-Chat](#) — *LangChain, ChromaDB, Groq (Llama 3.3 70B), FAISS, Sentence Transformers, Multi Agent system*

[GitHub](#)

- Engineered a hallucination-resistant RAG pipeline ingesting PDF/DOCX/PPTX/CSV files at 446K+ characters/second using locally generated embeddings for complete data privacy across a benchmarked, multi-stage retrieval architecture.
- Achieved 100% citation-backed responses with 0.69s average end-to-end query latency, ensuring fully source-cited, hallucination-resistant answers.

[CreditRisk](#) — *XGBoost, SHAP, FastAPI, Logistic Regression, Random Forest Classifier, Scikit-Learn, EDA, Pre-Processing*

[GitHub](#)

- Rectified XGBoost loan default classifier achieving 93.6% accuracy and 0.95 ROC-AUC on 32,000+ records, outperforming Logistic Regression and Random Forest baselines; developed SHAP-based explainability engine converting log-odds into human-readable risk factors using custom React waterfall chart visualization.

[Co-Drive](#) — *YOLO11n, FastAPI, , CUDA, Ultralytics, OpenCV, Fine-tuning, , EDA, Pre-Processing*

[GitHub](#)

- Fine-tuned a custom YOLO11n computer-vision model for real-time sign detection, reaching 99.05% mAP accuracy at 19–25ms GPU inference latency. Shipped a full-stack detection suite supporting image, video, and live webcam recognition through a unified React dashboard.

[AgriDiagnostix](#) — *PyTorch, EfficientNet-B0, Scikit-Learn, Decision Trees, Random Forest, SVM, XGBoost, CNN, MobileNetV3*

[GitHub](#)

- Trained and benchmarked multiple models (Logistic Regression, Decision Tree, Random Forest, XGBoost, CNN, MobileNetV3) for 16-class crop disease diagnosis, with EfficientNet-B0 achieving the best result (96.6% val. accuracy); shipped via FastAPI + React app enabling sub-2s live-camera diagnosis and treatment guidance for field use.

[MoodFlix](#) — *FastAPI, Grok LLM, Scikit-Learn, TF-IDF, Cosine Similarity, TMDB, Fuzzy Matching*

[GitHub](#)

- Constructed a mood-based movie recommendation engine combining TF-IDF vectorization, cosine similarity, and LLM-powered natural-language intent extraction, with an asynchronous FastAPI backend that reduced metadata-fetch latency by 70% through parallelized TMDB API calls.

CERTIFICATIONS

- [BCG X — GenAI Engineer](#) Job Simulation — Forage

[Certificate](#)

- [JPMorgan Chase & Co. — Quantitative Research Analyst](#) Job Simulation — Forage

[Certificate](#)